FILIPPO MARCANTONI

Robotics Software Engineer | Computer Vision & 3D Perception | Robot Learning | Physical AI

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SUMMARY

Robotics engineer with an M.S. in Robotics Engineering from Worcester Polytechnic Institute and hands-on R&D experience across robotics software, computer vision, robot learning, industrial automation, teleoperation, swarm robotics, and generative AI. Experienced in building and integrating systems across perception, simulation, learning, testing, and real-world deployment. Seeking robotics software, robot learning, computer vision, or physical AI roles with the goal of contributing to practical robotic systems and continuing to grow as a full-stack robotics engineer.

EDUCATION

Worcester Polytechnic Institute (WPI) — *Worcester, MA, USA*

M.S. Robotics Engineering | GPA 3.90/4.00 — *Aug 2024 – May 2026*

B.S. Robotics Engineering | GPA 3.89/4.00, Summa Cum Laude — *Aug 2021 – May 2025*

EXPERIENCE

AI & Industrial Robotics R&D Intern | B&R Industrial Automation, Austria — *Jun 2025 – Aug 2025*

- Developed an LLM-based code-generation copilot for B&R Automation Studio using Transformers, RAG, Azure AI Search, and domain-specific Motion Structured Text (Motion ST) examples to support industrial robotics and automation workflows.
- Designed the data-ingestion and retrieval backend using domain-aware metadata, custom syntax-aware chunking for Structured Text and Motion ST, and hybrid dense and sparse retrieval with PostgreSQL, pgvector, and BM25.

Machine Vision & AI R&D Intern | Makro Labelling SRL, Italy — *May 2024 – Aug 2024*

- Developed C++ and Python machine-vision pipelines for bottle orientation, label alignment, and defect inspection on high-speed labeling lines.
- Benchmarked classical and deep-learning vision models across bottle formats, lighting conditions, and label shifts to identify failure modes, validate robustness, and improve automated quality-control performance.

RESEARCH

PiELo: Reactive Swarm Programming Language | NEST Lab, WPI — *Aug 2024 – May 2025*

- Co-developed PiELo, a reactive C++ programming language for robot swarms, implementing core compiler and runtime components including a tokenizer, parser, AST-to-bytecode compiler, stack-based virtual machine, reactive closures, and mark-and-sweep garbage collection.
- Implemented distributed coordination mechanisms, including a last-writer-wins shared dictionary, and validated consensus, barrier, and leader-election behaviors in ARGoS3 and on Khepera IV robots with up to 90% packet loss.

VR-Based Multimodal Teleoperation Interface | HiRo Lab, WPI — *Jan 2025 – May 2025*

- Developed a human-in-the-loop VR teleoperation system for a robotic nursing assistant using Unity, ROS, Meta Quest head-motion mirroring, real-time multi-camera streaming, Whisper voice commands, and ZED-Mini 3D object detection.
- Integrated viewpoint control, speech-based camera switching, and perception-driven camera selection to improve operator situational awareness during remote robotic manipulation.

SELECTED PROJECTS

Robot Learning: Reinforcement & Imitation Learning

- Implemented policy iteration, REINFORCE, A2C, and A3C for simulated control and robotic manipulation tasks, including LunarLander and PyBullet Kuka vision-based grasping, reaching 55% grasp success with a CNN actor-critic policy.
- Trained imitation-learning policies in robomimic on 60 PickPlaceCan demonstrations, improving success from 0.02 with baseline behavioral cloning to 0.80 with LSTM behavioral cloning and 0.90 with diffusion policy.

Classical MSCKF & Deep Visual-Inertial Odometry

- Implemented a stereo MSCKF visual-inertial odometry backend with IMU propagation, camera-state cloning, stereo feature tracking, and nullspace EKF updates, evaluated on EuRoC and RPG datasets.
- Trained IMU-only, vision-only, and fused deep VIO networks on synthetic Blender trajectories, evaluating sensor-fusion behavior with geodesic SO(3) and trajectory-composition losses.

Monocular 3D Driving-Scene Reconstruction

- Built a monocular 3D perception pipeline integrating object detection, segmentation, depth estimation, pose estimation, and optical flow for autonomous-driving scene understanding.
- Reconstructed 3D driving scenes in Blender with lanes, vehicles, pedestrians, traffic objects, orientation, motion state, and collision-risk estimation.

TECHNICAL SKILLS

Languages: Python, C++, C, C#, MATLAB, Java, JavaScript, Buzz.

Robotics & Simulation: ROS/ROS2, Gazebo, MuJoCo, PyBullet, Unity, Blender, Vicon, VR/XR, Automation Studio.

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, TensorRT, CUDA, Transformers, LLMs, RAG, Reinforcement Learning, Imitation Learning, Behavioral Cloning, Diffusion Policy, Actor-Critic Methods.

Computer Vision & 3D Perception: OpenCV, SLAM, Visual-Inertial Odometry, MSCKF, SfM, NeRF, RANSAC, PnP, Bundle Adjustment, Homography, Object Detection, Segmentation, Monocular Depth, Pose Estimation, Optical Flow, Point Clouds.

Tools & Infrastructure: Linux, Git, Docker, Jupyter, Azure AI, AWS, Simulink, React, Node.js, PostgreSQL, MySQL, HPC/Slurm.

CAD & Design: SolidWorks, Fusion 360.